

TASK 1: Temperature Monitoring & LCD Display

1. Aim :

To measure the surrounding temperature using an **LM35 temperature sensor** and display the current temperature on a **16×2 I2C LCD** using an **Arduino UNO**. The temperature readings are also monitored through the Serial Monitor.

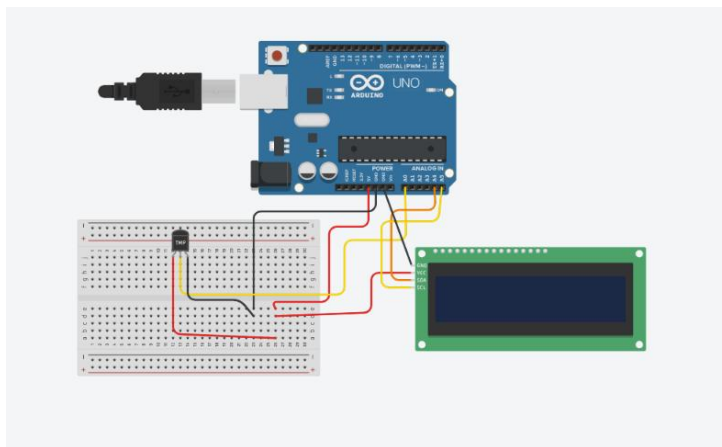
2. Goal :

Read the temperature from a sensor and display the current value on a 16×2 LCD. Log the readings through Serial communication.

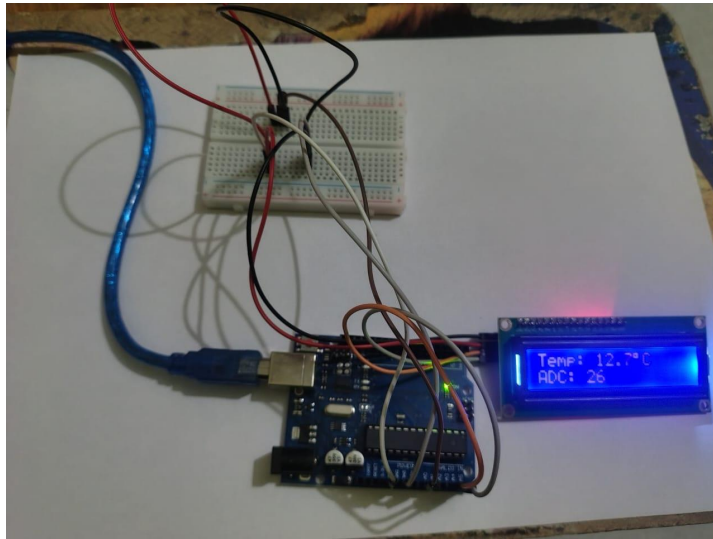
3. Components Required:

S. No	Components	Qty
1	Arduino UNO R3	1
2	LM35 Temperature Sensor	1
3	16×2 I2C LCD Display	1
4	Breadboard	1
5	Jumper Wires	As required
6	USB Cable	1

4. Circuit Connections:



5. Complete Circuit Connection:



6. Arduino Program:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD I2C address: 0x27
LiquidCrystal_I2C lcd(0x27, 16, 2);

// LM35 output connected to Analog Pin A0
const int sensorPin = A0;

void setup() {
  Serial.begin(9600);

  // Initialize LCD
  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print("Temperature");
  lcd.setCursor(0, 1);
  lcd.print("Monitoring");

  delay(2000);

  lcd.clear();
}
```

```
void loop() {  
  
    // Read analog value from LM35  
    int sensorValue = analogRead(sensorPin);  
  
    // Convert ADC value to voltage  
    float voltage = sensorValue * (5.0 / 1023.0);  
  
    // LM35: 10 mV per degree Celsius  
    float temperature = voltage * 100.0;  
  
    // Print temperature on LCD  
    lcd.setCursor(0, 0);  
    lcd.print("Temp: ");  
  
    lcd.print(temperature, 1);  
    lcd.print((char)223);  
    lcd.print("C  ");  
  
    // Print temperature on Serial Monitor  
    Serial.print("Temperature: ");  
    Serial.print(temperature, 1);  
    Serial.println(" C");  
  
    delay(1000);  
}
```

7. OUTPUT:

